

Integration by Parts

ANSWER KEY



The integration by parts formula

- 1 State the integration by parts formula, and explain the LIATE guideline for choosing u .

$\int u dv = uv - \int v du$. LIATE suggests choosing u in the priority order: Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — the function type listed first should typically be u .

- 2 Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.

Integrating both sides: $uv = \int u'v dx + \int uv' dx = \int v du + \int u dv$. Rearranging: $\int u dv = uv - \int v du$.

- 3 Evaluate $\int xe^x dx$.

Let $u = x$, $dv = e^x dx$: $du = dx$, $v = e^x$. $\int xe^x dx = xe^x - \int e^x dx = xe^x - e^x + C$.

- 4 Evaluate $\int x \cos x dx$.

Let $u = x$, $dv = \cos x dx$: $du = dx$, $v = \sin x$. $\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C$.

Integration by parts with logarithms and inverse trig

- 5 Evaluate $\int \ln x dx$.

Let $u = \ln x$, $dv = dx$: $du = \frac{1}{x} dx$, $v = x$. $\int \ln x dx = x \ln x - \int 1 dx = x \ln x - x + C$.

- 6 Evaluate $\int x \ln x dx$.

Let $u = \ln x$, $dv = x dx$: $du = \frac{1}{x} dx$, $v = \frac{x^2}{2}$. $\int x \ln x dx = \frac{x^2}{2} \ln x - \int \frac{x}{2} dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$.

- 7 Evaluate $\int \tan^{-1} x dx$.

Let $u = \tan^{-1} x$, $dv = dx$: $du = \frac{1}{1+x^2} dx$, $v = x$.
 $\int \tan^{-1} x dx = x \tan^{-1} x - \int \frac{x}{1+x^2} dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$.

- 8 Evaluate $\int x^2 \ln x dx$.

Let $u = \ln x$, $dv = x^2 dx$: $du = \frac{1}{x} dx$, $v = \frac{x^3}{3}$. $\int x^2 \ln x dx = \frac{x^3}{3} \ln x - \int \frac{x^2}{3} dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C$.

- 9 Evaluate $\int \sin^{-1} x dx$.

Let $u = \sin^{-1} x$, $dv = dx$: $du = \frac{1}{\sqrt{1-x^2}} dx$, $v = x$.
 $\int \sin^{-1} x dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx = x \sin^{-1} x + \sqrt{1-x^2} + C$.

Repeated integration by parts

- 10 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.

First: $u = x^2$, $dv = e^x dx$: $\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx$. Using the earlier result $\int x e^x dx = x e^x - e^x$:
 $= x^2 e^x - 2(x e^x - e^x) + C = x^2 e^x - 2x e^x + 2e^x + C$.

- 11 Evaluate $\int x^2 \sin x dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \sin x dx$: $= -x^2 \cos x + 2 \int x \cos x dx$. Using the earlier result
 $\int x \cos x dx = x \sin x + \cos x$: $= -x^2 \cos x + 2x \sin x + 2 \cos x + C$.

- 12 Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x dx$, identifying how many times integration by parts must be applied.

Each application of IBP with $u = x^n$ reduces the power of x by 1 while v stays an antiderivative of e^x . Since the power starts at 3, integration by parts must be applied 3 times to eliminate the polynomial factor entirely.

The cyclic case

- 13 Evaluate $\int e^x \sin x dx$ using the cyclic (solve-for-the-integral) technique.

Let $I = \int e^x \sin x dx$. First IBP ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x dx$. Second IBP on the remaining integral ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x dx = e^x \cos x + \int e^x \sin x dx = e^x \cos x + I$. So
 $I = e^x \sin x - (e^x \cos x + I) \Rightarrow 2I = e^x \sin x - e^x \cos x \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$.

- 14 Evaluate $\int e^{2x} \cos x dx$ using the cyclic technique.

Let $I = \int e^{2x} \cos x dx$. IBP ($u = \cos x$, $dv = e^{2x} dx$): $I = \frac{1}{2} e^{2x} \cos x + \frac{1}{2} \int e^{2x} \sin x dx$. IBP again on that integral ($u = \sin x$, $dv = e^{2x} dx$): $\int e^{2x} \sin x dx = \frac{1}{2} e^{2x} \sin x - \frac{1}{2} I$. Substituting:
 $I = \frac{1}{2} e^{2x} \cos x + \frac{1}{4} e^{2x} \sin x - \frac{1}{4} I \Rightarrow \frac{5}{4} I = \frac{1}{2} e^{2x} \cos x + \frac{1}{4} e^{2x} \sin x \Rightarrow I = \frac{e^{2x}(2 \cos x + \sin x)}{5} + C$.

Definite integrals using integration by parts

- 15 Evaluate $\int_0^1 x e^x dx$ using the antiderivative found earlier.

$$[x e^x - e^x]_0^1 = (e - e) - (0 - 1) = 0 - (-1) = 1.$$

- 16 Evaluate $\int_1^e \ln x dx$ using the antiderivative found earlier.

$$[x \ln x - x]_1^e = (e \cdot 1 - e) - (0 - 1) = 0 - (-1) = 1.$$

- 17 Evaluate $\int_0^\pi x \sin x dx$.

$$\text{Let } u = x, dv = \sin x dx: \int x \sin x dx = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C.$$
$$[-x \cos x + \sin x]_0^\pi = (-\pi(-1) + 0) - (0 + 0) = \pi.$$

Integration by parts to derive a reduction formula

- 18** Use integration by parts to show $\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x dx$.

Let $u = x^n$, $dv = e^x dx$: $du = nx^{n-1} dx$, $v = e^x$. $\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$, confirming the reduction formula. With $n = 2$: $\int x^2 e^x dx = x^2 e^x - 2 \int x e^x dx = x^2 e^x - 2x e^x + 2e^x + C$, matching the earlier result.

Choosing u and dv wisely

- 19** For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x dx$ $u = \ln x$ (Logarithmic beats Algebraic) **b** $\int x e^{5x} dx$ $u = x$ (Algebraic beats Exponential)

c $\int x \tan^{-1} x dx$ $u = \tan^{-1} x$ (Inverse trig beats Algebraic) **d** $\int x^2 \cos x dx$ $u = x^2$ (Algebraic beats Trigonometric)

- 20** Explain what goes wrong if you attempt $\int x e^x dx$ with the reversed choice $u = e^x$, $dv = x dx$.

Then $du = e^x dx$, $v = \frac{x^2}{2}$, giving $\int x e^x dx = \frac{x^2}{2} e^x - \int \frac{x^2}{2} e^x dx$. The remaining integral is MORE complicated than the original (higher power of x under the same exponential), so this choice makes the problem worse rather than simpler — LIATE's algebraic-before-exponential guidance avoids this trap.

Integration by parts with algebraic-trigonometric products

- 21** Evaluate $\int x \sec^2 x dx$.

Let $u = x$, $dv = \sec^2 x dx$: $du = dx$, $v = \tan x$.
 $\int x \sec^2 x dx = x \tan x - \int \tan x dx = x \tan x + \ln|\cos x| + C$.

- 22** Evaluate $\int 2x \sin(3x) dx$.

Let $u = 2x$, $dv = \sin(3x) dx$: $du = 2 dx$, $v = -\frac{1}{3} \cos(3x)$.
 $\int 2x \sin(3x) dx = -\frac{2x}{3} \cos(3x) + \frac{2}{3} \int \cos(3x) dx = -\frac{2x}{3} \cos(3x) + \frac{2}{9} \sin(3x) + C$.

Definite integrals: area under a transcendental product

- 23 Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x} dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).

Check: $\frac{d}{dx}[-xe^{-x} - e^{-x}] = -e^{-x} + xe^{-x} + e^{-x} = xe^{-x} \checkmark$. Area
 $= [-xe^{-x} - e^{-x}]_0^3 = (-3e^{-3} - e^{-3}) - (0 - 1) = 1 - 4e^{-3} \approx 1 - 0.199 \approx 0.801$.

- 24 Evaluate $\int_0^{\pi/2} x \cos x dx$ using the antiderivative $x \sin x + \cos x$ found earlier.

$$[x \sin x + \cos x]_0^{\pi/2} = \left(\frac{\pi}{2}(1) + 0\right) - (0 + 1) = \frac{\pi}{2} - 1 \approx 0.571.$$

Mixed review: choosing the right technique

- 25 For each integral, state whether it should be solved by u -substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:

a $\int 2xe^{x^2} dx$ Substitution ($w = x^2$) — the derivative of the exponent is present as a factor

b $\int x^2 e^x dx$ Integration by parts ($u = x^2$, twice) — no matching derivative factor for substitution

c $\int \frac{\ln x}{x} dx$ Substitution ($w = \ln x$) — $\frac{1}{x}$ is exactly dw

d $\int x \ln x dx$ Integration by parts ($u = \ln x$) — no simplifying substitution exists here

Multiple choice — easy 15

- 26 Evaluate $\int_0^{\pi} -2x \cos x dx$, using integration by parts.

- a. 2
- b. 6
- c. 4 ✓
- d. -4

Let $u = -2x$, $dv = \cos x dx$: $du = -2 dx$, $v = \sin x$. $\int -2x \cos x dx = -2x \sin x - 2 \cos x + C$.
Evaluating: $[-2x \sin x - 2 \cos x]_0^{\pi} = (0 + 2) - (0 - 2) = 4$.

27 Evaluate $\int_0^{\pi} 3x \cos x \, dx$, using integration by parts.

a. -6 ✓

b. -3

c. 6

d. -4

Let $u = 3x$, $dv = \cos x \, dx$: $du = 3 \, dx$, $v = \sin x$. $\int 3x \cos x \, dx = 3x \sin x + 3 \cos x + C$. Evaluating:
 $[3x \sin x + 3 \cos x]_0^{\pi} = (0 - 3) - (0 + 3) = -6$.

28 Evaluate $\int_0^{\pi} -6x \cos x \, dx$, using integration by parts.

a. 6

b. 14

c. 12 ✓

d. -12

Let $u = -6x$, $dv = \cos x \, dx$: $du = -6 \, dx$, $v = \sin x$. $\int -6x \cos x \, dx = -6x \sin x - 6 \cos x + C$.
Evaluating: $[-6x \sin x - 6 \cos x]_0^{\pi} = (0 + 6) - (0 - 6) = 12$.

29 Evaluate $\int_0^{\pi} x \cos x \, dx$, using integration by parts.

a. 2

b. 0

c. -1

d. -2 ✓

Let $u = x$, $dv = \cos x \, dx$: $du = 1 \, dx$, $v = \sin x$. $\int x \cos x \, dx = x \sin x + \cos x + C$. Evaluating:
 $[x \sin x + \cos x]_0^{\pi} = (0 - 1) - (0 + 1) = -2$.

30 Evaluate $\int_0^{\pi} -7x \cos x \, dx$, using integration by parts.

- a. 16
- b. 7
- c. -14
- d. 14 ✓

Let $u = -7x$, $dv = \cos x \, dx$: $du = -7 \, dx$, $v = \sin x$. $\int -7x \cos x \, dx = -7x \sin x - 7 \cos x + C$.
Evaluating: $[-7x \sin x - 7 \cos x]_0^{\pi} = (0 + 7) - (0 - 7) = 14$.

31 Evaluate $\int_0^{\pi} 7x \cos x \, dx$, using integration by parts.

- a. -14 ✓
- b. -12
- c. -7
- d. 14

Let $u = 7x$, $dv = \cos x \, dx$: $du = 7 \, dx$, $v = \sin x$. $\int 7x \cos x \, dx = 7x \sin x + 7 \cos x + C$. Evaluating:
 $[7x \sin x + 7 \cos x]_0^{\pi} = (0 - 7) - (0 + 7) = -14$.

32 Evaluate $\int_0^{\pi} 9x \cos x \, dx$, using integration by parts.

- a. -16
- b. 18
- c. -18 ✓
- d. -9

Let $u = 9x$, $dv = \cos x \, dx$: $du = 9 \, dx$, $v = \sin x$. $\int 9x \cos x \, dx = 9x \sin x + 9 \cos x + C$. Evaluating:
 $[9x \sin x + 9 \cos x]_0^{\pi} = (0 - 9) - (0 + 9) = -18$.

33 Evaluate $\int_0^{\pi} -8x \cos x \, dx$, using integration by parts.

a. 18

b. 16 ✓

c. -16

d. 8

Let $u = -8x$, $dv = \cos x \, dx$: $du = -8 \, dx$, $v = \sin x$. $\int -8x \cos x \, dx = -8x \sin x - 8 \cos x + C$.
Evaluating: $[-8x \sin x - 8 \cos x]_0^{\pi} = (0 + 8) - (0 - 8) = 16$.

34 Evaluate $\int_0^{\pi} 8x \cos x \, dx$, using integration by parts.

a. -16 ✓

b. 16

c. -14

d. -8

Let $u = 8x$, $dv = \cos x \, dx$: $du = 8 \, dx$, $v = \sin x$. $\int 8x \cos x \, dx = 8x \sin x + 8 \cos x + C$. Evaluating:
 $[8x \sin x + 8 \cos x]_0^{\pi} = (0 - 8) - (0 + 8) = -16$.

35 Evaluate $\int_0^{\pi} -x \cos x \, dx$, using integration by parts.

a. -2

b. 4

c. 2 ✓

d. 1

Let $u = -x$, $dv = \cos x \, dx$: $du = -1 \, dx$, $v = \sin x$. $\int -x \cos x \, dx = -x \sin x - \cos x + C$. Evaluating:
 $[-x \sin x - \cos x]_0^{\pi} = (0 + 1) - (0 - 1) = 2$.

36 Evaluate $\int_0^{\pi} 6x \cos x \, dx$, using integration by parts.

- a. 12
- b. -12 ✓
- c. -10
- d. -6

Let $u = 6x$, $dv = \cos x \, dx$: $du = 6 \, dx$, $v = \sin x$. $\int 6x \cos x \, dx = 6x \sin x + 6 \cos x + C$. Evaluating: $[6x \sin x + 6 \cos x]_0^{\pi} = (0 - 6) - (0 + 6) = -12$.

37 Evaluate $\int_0^{\pi} -3x \cos x \, dx$, using integration by parts.

- a. 3
- b. -6
- c. 8
- d. 6 ✓

Let $u = -3x$, $dv = \cos x \, dx$: $du = -3 \, dx$, $v = \sin x$. $\int -3x \cos x \, dx = -3x \sin x - 3 \cos x + C$. Evaluating: $[-3x \sin x - 3 \cos x]_0^{\pi} = (0 + 3) - (0 - 3) = 6$.

38 Evaluate $\int_0^{\pi} -9x \cos x \, dx$, using integration by parts.

- a. 9
- b. -18
- c. 20
- d. 18 ✓

Let $u = -9x$, $dv = \cos x \, dx$: $du = -9 \, dx$, $v = \sin x$. $\int -9x \cos x \, dx = -9x \sin x - 9 \cos x + C$. Evaluating: $[-9x \sin x - 9 \cos x]_0^{\pi} = (0 + 9) - (0 - 9) = 18$.

39 Evaluate $\int_0^{\pi} -4x \cos x \, dx$, using integration by parts.

- a. 4
- b. -8
- c. 10
- d. 8 ✓

Let $u = -4x$, $dv = \cos x \, dx$: $du = -4 \, dx$, $v = \sin x$. $\int -4x \cos x \, dx = -4x \sin x - 4 \cos x + C$.
Evaluating: $[-4x \sin x - 4 \cos x]_0^{\pi} = (0 + 4) - (0 - 4) = 8$.

40 Evaluate $\int_0^{\pi} 5x \cos x \, dx$, using integration by parts.

- a. -8
- b. 10
- c. -10 ✓
- d. -5

Let $u = 5x$, $dv = \cos x \, dx$: $du = 5 \, dx$, $v = \sin x$. $\int 5x \cos x \, dx = 5x \sin x + 5 \cos x + C$. Evaluating:
 $[5x \sin x + 5 \cos x]_0^{\pi} = (0 - 5) - (0 + 5) = -10$.

Multiple choice — medium 15

41 Given $f(0) = 0$, $f(3) = 2$, $g(0) = 2$, $g(3) = 2$, and $\int_0^3 f'(x)g(x) \, dx = -2$, use integration by parts to find $\int_0^3 f(x)g'(x) \, dx$.

- a. 10
- b. -6
- c. 2
- d. 6 ✓

$$\int_0^3 f(x)g'(x) \, dx = [f(x)g(x)]_0^3 - \int_0^3 f'(x)g(x) \, dx = [(2)(2) - (0)(2)] - (-2) = 4 - (-2) = 6.$$

- 42 Given $f(0) = 0$, $f(1) = 2$, $g(0) = 3$, $g(1) = 5$, and $\int_0^1 f'(x)g(x) dx = 7$, use integration by parts to find $\int_0^1 f(x)g'(x) dx$.

- a. 7
- b. -3
- c. 17
- d. 3 ✓

$$\int_0^1 f(x)g'(x) dx = [f(x)g(x)]_0^1 - \int_0^1 f'(x)g(x) dx = [(2)(5) - (0)(3)] - (7) = 10 - (7) = 3.$$

- 43 Given $f(2) = 4$, $f(7) = 4$, $g(2) = 0$, $g(7) = -2$, and $\int_2^7 f'(x)g(x) dx = -14$, use integration by parts to find $\int_2^7 f(x)g'(x) dx$.

- a. 10
- b. -22
- c. -6
- d. 6 ✓

$$\int_2^7 f(x)g'(x) dx = [f(x)g(x)]_2^7 - \int_2^7 f'(x)g(x) dx = [(4)(-2) - (4)(0)] - (-14) = -8 - (-14) = 6.$$

- 44 Given $f(1) = 5$, $f(3) = 6$, $g(1) = 4$, $g(3) = -6$, and $\int_1^3 f'(x)g(x) dx = -7$, use integration by parts to find $\int_1^3 f(x)g'(x) dx$.

- a. -45
- b. -49 ✓
- c. -63
- d. 49

$$\int_1^3 f(x)g'(x) dx = [f(x)g(x)]_1^3 - \int_1^3 f'(x)g(x) dx = [(6)(-6) - (5)(4)] - (-7) = -56 - (-7) = -49.$$

- 45 Given $f(0) = -5$, $f(4) = 6$, $g(0) = -1$, $g(4) = 1$, and $\int_0^4 f'(x)g(x) dx = -12$, use integration by parts to find $\int_0^4 f(x)g'(x) dx$.

- a. -13
- b. -11
- c. 13 ✓
- d. 17

$$\int_0^4 f(x)g'(x) dx = [f(x)g(x)]_0^4 - \int_0^4 f'(x)g(x) dx = [(6)(1) - (-5)(-1)] - (-12) = 1 - (-12) = 13.$$

- 46 Given $f(3) = -1$, $f(6) = 2$, $g(3) = 1$, $g(6) = 3$, and $\int_3^6 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^6 f(x)g'(x) dx$.

- a. -1
- b. 15 ✓
- c. -15
- d. 19

$$\int_3^6 f(x)g'(x) dx = [f(x)g(x)]_3^6 - \int_3^6 f'(x)g(x) dx = [(2)(3) - (-1)(1)] - (-8) = 7 - (-8) = 15.$$

- 47 Given $f(3) = -2$, $f(6) = 6$, $g(3) = 6$, $g(6) = 3$, and $\int_3^6 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^6 f(x)g'(x) dx$.

- a. 38 ✓
- b. 42
- c. -38
- d. 22

$$\int_3^6 f(x)g'(x) dx = [f(x)g(x)]_3^6 - \int_3^6 f'(x)g(x) dx = [(6)(3) - (-2)(6)] - (-8) = 30 - (-8) = 38.$$

- 48 Given $f(3) = 5$, $f(7) = 0$, $g(3) = 6$, $g(7) = -5$, and $\int_3^7 f'(x)g(x) dx = -10$, use integration by parts to find $\int_3^7 f(x)g'(x) dx$.

a. -40

b. -20 ✓

c. 20

d. -16

$$\int_3^7 f(x)g'(x) dx = [f(x)g(x)]_3^7 - \int_3^7 f'(x)g(x) dx = [(0)(-5) - (5)(6)] - (-10) = -30 - (-10) = -20.$$

- 49 Given $f(0) = -1$, $f(5) = 4$, $g(0) = -2$, $g(5) = -3$, and $\int_0^5 f'(x)g(x) dx = 11$, use integration by parts to find $\int_0^5 f(x)g'(x) dx$.

a. 25

b. -21

c. -3

d. -25 ✓

$$\int_0^5 f(x)g'(x) dx = [f(x)g(x)]_0^5 - \int_0^5 f'(x)g(x) dx = [(4)(-3) - (-1)(-2)] - (11) = -14 - (11) = -25.$$

- 50 Given $f(0) = 1$, $f(4) = 1$, $g(0) = 4$, $g(4) = 0$, and $\int_0^4 f'(x)g(x) dx = -5$, use integration by parts to find $\int_0^4 f(x)g'(x) dx$.

a. -1

b. 1 ✓

c. -9

d. 5

$$\int_0^4 f(x)g'(x) dx = [f(x)g(x)]_0^4 - \int_0^4 f'(x)g(x) dx = [(1)(0) - (1)(4)] - (-5) = -4 - (-5) = 1.$$

- 51 Given $f(0) = 5$, $f(3) = -2$, $g(0) = 3$, $g(3) = 1$, and $\int_0^3 f'(x)g(x) dx = -6$, use integration by parts to find $\int_0^3 f(x)g'(x) dx$.

- a. -7
- b. -11 ✓
- c. -23
- d. 11

$$\int_0^3 f(x)g'(x) dx = [f(x)g(x)]_0^3 - \int_0^3 f'(x)g(x) dx = [(-2)(1) - (5)(3)] - (-6) = -17 - (-6) = -11.$$

- 52 Given $f(1) = -5$, $f(5) = -3$, $g(1) = 3$, $g(5) = -1$, and $\int_1^5 f'(x)g(x) dx = 4$, use integration by parts to find $\int_1^5 f(x)g'(x) dx$.

- a. 18
- b. 22
- c. 14 ✓
- d. -14

$$\int_1^5 f(x)g'(x) dx = [f(x)g(x)]_1^5 - \int_1^5 f'(x)g(x) dx = [(-3)(-1) - (-5)(3)] - (4) = 18 - (4) = 14.$$

- 53 Given $f(3) = -4$, $f(4) = 6$, $g(3) = -4$, $g(4) = -6$, and $\int_3^4 f'(x)g(x) dx = -2$, use integration by parts to find $\int_3^4 f(x)g'(x) dx$.

- a. -46
- b. -50 ✓
- c. -54
- d. 50

$$\int_3^4 f(x)g'(x) dx = [f(x)g(x)]_3^4 - \int_3^4 f'(x)g(x) dx = [(6)(-6) - (-4)(-4)] - (-2) = -52 - (-2) = -50.$$

- 54 Given $f(3) = -6$, $f(7) = 5$, $g(3) = 0$, $g(7) = 0$, and $\int_3^7 f'(x)g(x) dx = -8$, use integration by parts to find $\int_3^7 f(x)g'(x) dx$.

- a. 12
- b. -8
- c. 8 ✓
- d. 15

$$\int_3^7 f(x)g'(x) dx = [f(x)g(x)]_3^7 - \int_3^7 f'(x)g(x) dx = [(5)(0) - (-6)(0)] - (-8) = 0 - (-8) = 8.$$

- 55 Given $f(0) = 0$, $f(3) = -6$, $g(0) = -3$, $g(3) = 4$, and $\int_0^3 f'(x)g(x) dx = -4$, use integration by parts to find $\int_0^3 f(x)g'(x) dx$.

- a. -28
- b. 20
- c. -20 ✓
- d. -16

$$\int_0^3 f(x)g'(x) dx = [f(x)g(x)]_0^3 - \int_0^3 f'(x)g(x) dx = [(-6)(4) - (0)(-3)] - (-4) = -24 - (-4) = -20.$$

Multiple choice — hard 15

- 56 Using the cyclic technique, $\int e^x \cos(2x) dx = \frac{e^x(\cos(2x) + C_1 \sin(2x))}{5} + C$. Find the coefficient C_1 .

- a. $\frac{1}{5}$
- b. $-\frac{2}{5}$
- c. 2
- d. $\frac{2}{5}$ ✓

By the standard cyclic-IBP result, $\int e^{ax} \cos(bx) dx = \frac{e^{ax}(a \cos(bx) + b \sin(bx))}{a^2 + b^2} + C$. With $a = 1$, $b = 2$: $C_1 = b = 2$, and the denominator is $1^2 + 2^2 = 5$, so as a fraction of the denominator, $C_1 = \frac{2}{5} = \frac{2}{5}$.

57 Using the cyclic technique, $\int e^{3x} \cos(3x) dx = \frac{e^{3x}(3\cos(3x) + C_1 \sin(3x))}{18} + C$. Find the coefficient C_1 .

- a. $\frac{1}{3}$
- b. $-\frac{1}{6}$
- c. $\frac{1}{6}$ ✓
- d. $\frac{7}{6}$

By the standard cyclic-IBP result, $\int e^{ax} \cos(bx) dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 3$, $b = 3$: $C_1 = b = 3$, and the denominator is $3^2 + 3^2 = 18$, so as a fraction of the denominator, $C_1 = \frac{3}{18} = \frac{1}{6}$.

58 Using the cyclic technique, $\int e^{3x} \cos(5x) dx = \frac{e^{3x}(3\cos(5x) + C_1 \sin(5x))}{34} + C$. Find the coefficient C_1 .

- a. $\frac{5}{9}$
- b. $\frac{5}{34}$ ✓
- c. $-\frac{5}{34}$
- d. $\frac{3}{34}$

By the standard cyclic-IBP result, $\int e^{ax} \cos(bx) dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 3$, $b = 5$: $C_1 = b = 5$, and the denominator is $3^2 + 5^2 = 34$, so as a fraction of the denominator, $C_1 = \frac{5}{34} = \frac{5}{34}$.

59 Using the cyclic technique, $\int e^x \cos(5x) dx = \frac{e^x(\cos(5x) + C_1 \sin(5x))}{26} + C$. Find the coefficient C_1 .

- a. 5
- b. $\frac{1}{26}$
- c. $-\frac{5}{26}$
- d. $\frac{5}{26}$ ✓

By the standard cyclic-IBP result, $\int e^{ax} \cos(bx) dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 1$, $b = 5$: $C_1 = b = 5$, and the denominator is $1^2 + 5^2 = 26$, so as a fraction of the denominator, $C_1 = \frac{5}{26} = \frac{5}{26}$.

60 Using the cyclic technique, $\int e^{5x} \cos(4x) dx = \frac{e^{5x}(5\cos(4x) + C_1 \sin(4x))}{41} + C$. Find the coefficient C_1 .

- a. $-\frac{4}{41}$
- b. $\frac{4}{41}$ ✓
- c. $\frac{5}{41}$
- d. $\frac{4}{25}$

By the standard cyclic-IBP result, $\int e^{ax} \cos(bx) dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 5$, $b = 4$: $C_1 = b = 4$, and the denominator is $5^2 + 4^2 = 41$, so as a fraction of the denominator, $C_1 = \frac{4}{41} = \frac{4}{41}$.

61 For $\int x \ln x dx$, which is the correct choice of u according to LIATE?

- a. It does not matter which factor is chosen for u
- b. $u = x$ (Algebraic always goes first)
- c. $u = x$ (the opposite LIATE priority)
- d. $u = \ln x$ (Logarithmic outranks Algebraic in LIATE) ✓

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u : $u = \ln x$ (Logarithmic outranks Algebraic in LIATE).

62 For $\int x^2 \tan^{-1} x dx$, which is the correct choice of u according to LIATE?

- a. $u = \tan^{-1} x$ (Inverse Trig outranks Algebraic in LIATE) ✓
- b. $u = x^2$ (Algebraic always goes first)
- c. It does not matter which factor is chosen for u
- d. $u = x^2$ (the opposite LIATE priority)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u : $u = \tan^{-1} x$ (Inverse Trig outranks Algebraic in LIATE).

63 For $\int x^3 \ln x \, dx$, which is the correct choice of u according to LIATE?

- a. It does not matter which factor is chosen for u
- b. $u = x^3$ (the opposite LIATE priority)
- c. $u = \ln x$ (Logarithmic outranks Algebraic in LIATE) ✓
- d. $u = x^3$ (Algebraic always goes first)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u : $u = \ln x$ (Logarithmic outranks Algebraic in LIATE).

64 For $\int x^2 e^{6x} \, dx$, which is the correct choice of u according to LIATE?

- a. $u = x^2$ (Algebraic outranks Exponential in LIATE) ✓
- b. $u = x^2$ (Algebraic always goes first)
- c. It does not matter which factor is chosen for u
- d. $u = e^{6x}$ (the opposite LIATE priority)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u : $u = x^2$ (Algebraic outranks Exponential in LIATE).

65 For $\int x^3 \sin(6x) \, dx$, which is the correct choice of u according to LIATE?

- a. $u = \sin(6x)$ (the opposite LIATE priority)
- b. It does not matter which factor is chosen for u
- c. $u = x^3$ (Algebraic outranks Trigonometric in LIATE) ✓
- d. $u = x^3$ (Algebraic always goes first)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u : $u = x^3$ (Algebraic outranks Trigonometric in LIATE).

- 66 Given $f(2) = 5$, $f(5) = 2$, $g(2) = 5$, $g(5) = 2$, and $\int_2^5 f(x)g'(x) dx = -1$, use integration by parts to find $\int_2^5 f'(x)g(x) dx$.

a. -22

b. -20 ✓

c. -16

d. 20

Since $\int_2^5 f(x)g'(x) dx = [f(x)g(x)]_2^5 - \int_2^5 f'(x)g(x) dx$, rearranging:
 $\int_2^5 f'(x)g(x) dx = [(2)(2) - (5)(5)] - (-1) = -21 - (-1) = -20$.

- 67 Given $f(1) = 6$, $f(4) = 5$, $g(1) = 4$, $g(4) = 0$, and $\int_1^4 f(x)g'(x) dx = -14$, use integration by parts to find $\int_1^4 f'(x)g(x) dx$.

a. -6

b. -38

c. 10

d. -10 ✓

Since $\int_1^4 f(x)g'(x) dx = [f(x)g(x)]_1^4 - \int_1^4 f'(x)g(x) dx$, rearranging:
 $\int_1^4 f'(x)g(x) dx = [(5)(0) - (6)(4)] - (-14) = -24 - (-14) = -10$.

- 68 Given $f(1) = 5$, $f(3) = -1$, $g(1) = 2$, $g(3) = 0$, and $\int_1^3 f(x)g'(x) dx = -6$, use integration by parts to find $\int_1^3 f'(x)g(x) dx$.

a. 0

b. 4

c. -4 ✓

d. -16

Since $\int_1^3 f(x)g'(x) dx = [f(x)g(x)]_1^3 - \int_1^3 f'(x)g(x) dx$, rearranging:
 $\int_1^3 f'(x)g(x) dx = [(-1)(0) - (5)(2)] - (-6) = -10 - (-6) = -4$.

- 69 Given $f(0) = 2$, $f(2) = -2$, $g(0) = 0$, $g(2) = -3$, and $\int_0^2 f(x)g'(x) dx = -10$, use integration by parts to find $\int_0^2 f'(x)g(x) dx$.

- a. -4
- b. 16 ✓
- c. 20
- d. -16

Since $\int_0^2 f(x)g'(x) dx = [f(x)g(x)]_0^2 - \int_0^2 f'(x)g(x) dx$, rearranging:
 $\int_0^2 f'(x)g(x) dx = [(-2)(-3) - (2)(0)] - (-10) = 6 - (-10) = 16$.

- 70 Given $f(1) = 0$, $f(4) = 6$, $g(1) = -5$, $g(4) = -6$, and $\int_1^4 f(x)g'(x) dx = 1$, use integration by parts to find $\int_1^4 f'(x)g(x) dx$.

- a. 37
- b. -37 ✓
- c. -35
- d. -33

Since $\int_1^4 f(x)g'(x) dx = [f(x)g(x)]_1^4 - \int_1^4 f'(x)g(x) dx$, rearranging:
 $\int_1^4 f'(x)g(x) dx = [(6)(-6) - (0)(-5)] - (1) = -36 - (1) = -37$.